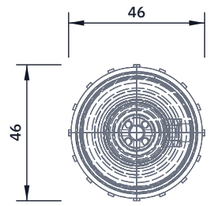
CONCEPT, NOT FOR FABRICATION

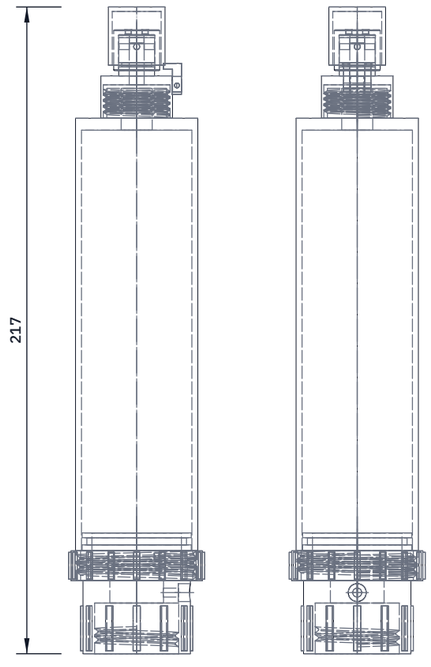
Helpful Water B · github.com/BoujeeEnjinia1701/helpful-water

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA from the TRL 3 model (HWB-CAL-001)	2026-11-11	AC



TOP VIEW

Scale 1:2; looking down (along -Z)



FRONT VIEW

Scale 1:2; looking at the front (along +Y)

RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; outside parts; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS (MM) AND DATA

Overall 217 long x 44 across the grip band
Fits 28 mm PCO 1881 and PCO 1810 necks; 38 mm by adapter
Housing 41 OD, 37 bore; 40 x 3 thread to the collar
Module 400 looped fibres 0.8/0.5, 0.17 m2, 0.1 um
Carbon capsule 30 x 20, about 7 g coconut GAC
Flow (estimate, clean): 465 mL/min at 17 kPa
Weak child 6 kPa: 170 mL/min (estimate)
Dry mass about 118 g (estimate)
Duckbill air valve in the collar; push-pull valve and dust cover
Backflush: 60 mL syringe at the nozzle, capsule removed
Not a virus barrier; co-treat with chlorine where viruses are a risk
PRELIMINARY, NOT FOR FABRICATION

HELPFUL WATER B

github.com/BoujeeEnjinia1701/helpful-water

B1 squeeze cartridge on a 28 mm PCO 1881 bottle neck: general arrangement

DRAWING NO.	REV	SHEET	SIZE
HWB-DWG-001	P1	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
1:2	mm		2026-11-11	Amish Chadha

MATERIAL / NOTES

PP housing, collar, capsule and spout; EPDM or silicone seals; 316L mesh; bought hollow-fibre module. See bom/bom.csv

Licensed CERN-OHL-P-2.0 · Design Molecule Labs with Helpful Engineering

Generated from cad/src; do not edit by hand